

$$2x^2 - 6x + 4 = 0$$
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$$x^2 - 3x + 2 = 0 \quad x=1$$

$$(x-1)(x-2) = 0 \quad x=2$$

Name

■ For the ODE  $2y'' - 6y' + 4y = 4\cos(2t)$

■ Write down the complementary sol.

$$y_c(t) = c_1 e^t + c_2 e^{2t}$$

■ Compute a particular sol.

$$y_p(t) = -\frac{1}{10} \cos(2t) - \frac{3}{10} \sin(2t)$$

■ The Aug Mat for  $y(0) = 1, y'(0) = 2$  is

~~Aug~~ 
$$\left[ \begin{array}{cc|c} 1 & 1 & 1 + \frac{1}{10} \\ 1 & 2 & 2 + \frac{6}{10} \end{array} \right]$$

1. For the ODE  $3y'' - 3y' - 18y = -3e^{4t}$

1.1. Write down the complementary sol.

$$y_c(t) =$$

1.2. Compute a particular sol.

$$y_p(t) =$$

1.3. The Aug Mat for  $y(0) = 2, y'(0) = 3$  is

$$y_p = a \cos(2t) + b \sin(2t)$$

$$y_p' = -2a \sin(2t) + 2b \cos(2t)$$

$$y_p'' = -4a \cos(2t) - 4b \sin(2t)$$

$$y_p'' - 3y_p' + 2y_p = 2 \cos(2t) \Rightarrow$$

$$\begin{aligned} & -4a \cos(2t) - 4b \sin(2t) \\ & -6b \cos(2t) + 6a \sin(2t) \\ & + 2a \cos(2t) + 2b \sin(2t) \\ & \hline 2 \cos(2t) + 0 \sin(2t) \end{aligned}$$

$$\begin{aligned} -2a - 6b &= 2 \\ 6a - 2b &= 0 \end{aligned}$$

$$\begin{aligned} -a - 3b &= 1 \\ 3a - b &= 0 \end{aligned}$$

$$b = 3a$$

$$-a - 3(3a) = 1$$

$$a = -\frac{1}{10} \quad b = -\frac{3}{10}$$

$$\begin{aligned} c_1 + c_2 - \frac{1}{10} &= 1 \\ c_1 + 2c_2 - \frac{6}{10} &= 2 \end{aligned}$$

$$y'' - y' - 6y = 0$$

~~$$y'' - y' - 6y = 0$$~~

$$x^2 - x - 6 = 0$$

$$x = -2 \text{ or } 3$$

$$y_p = a_2 t^2 + a_1 t + a_0$$

$$y_p' = 2a_2 t + a_1$$

$$y_p'' = 2a_2$$

$$y_p'' - y_p' - 6y_p = t^2 + \frac{1}{2}$$

$$\Rightarrow$$

- For the ODE  $2y'' - 2y' - 12y = 2t^2 + 1$  with

- Write down the complementary sol.

$$y_c(t) = c_1 e^{-2t} + c_2 e^{3t}$$

- 1.4. Give a  $y_p$  form and Aug Mat for the coefficients

$$y_p(t) = a_2 t^2 + a_1 t + a_0$$

$$\text{Aug Mat} = \begin{bmatrix} 0 & 0 & -6 & 1/2 \\ 0 & -6 & -2 & 0 \\ -6 & -1 & 2 & 1/2 \end{bmatrix}$$

$$\begin{array}{r} 2a_2 - a_1 - 6a_0 - 6a_2 t - 6a_2 t^2 \\ \hline 1/2 + 0t + 1/2 t^2 \end{array}$$

$$\begin{array}{r} 1/2 - 6a_0 - a_1 + 2a_2 = 1/2 \\ -6a_0 - 6a_1 - 2a_2 = 0 \\ +0a_0 - 6a_1 - 6a_2 = 1/2 \\ 0a_0 + 0a_1 - 6a_2 = 1/2 \end{array}$$

$$\begin{bmatrix} 0 & 0 & -6 \\ 0 & -6 & -2 \\ -6 & -1 & 2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 0 \\ 1/2 \end{bmatrix}$$

2. For the ODE  $2y'' + 2y' - 24y = 2\cos(3t)$

- 2.1. Write down the complementary sol.

$$y_c(t) =$$

- 2.2. Give a  $y_p$  form and Aug Mat for the coefficients

$$y_p(t) =$$

$$\text{Aug Mat} =$$